

Contest Mindset and Diagnostic

Entrance Diagnostic · Form A · Marking Key

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Answers and points

Q	Strand	Points	Answer	Q	Strand	Points	Answer
1	N	2	(D) 27	14	A	5	(B) 61
2	N	2	(B) 12	15	G	5	(C) 68
3	A	2	(C) 9	16	G	5	(B) 42
4	A	2	(D) 68	17	G	5	(C) 5 cm
5	G	2	(C) 38 cm	18	D	5	(D) 6
6	G	2	(C) 65°	19	T	5	(C) 47
7	D	2	(D) 14	20	L	5	(D) Dee
8	T	2	(A) 59	21	N	6	48
9	T	2	(E) 24	22	A	6	38
10	L	2	(A) Ben	23	G	6	24
11	N	5	(B) \$67.80	24	D	6	5/36
12	N	5	(B) 12	25	T	6	15
13	N	5	(C) $\frac{1}{2}$				

Strand codes: **N** Number, fractions, percent, ratio · **A** Algebra and patterns · **G** Geometry and measurement · **D** Data and probability · **T** Number theory and counting · **L** Logic. Blueprint: N 6 · A 4 · G 6 · D 3 · T 4 · L 2 items.

Figure notes

Q15 and Q16 are drawn to scale. Q15's missing sides (4 cm and 3 cm) are not labelled on purpose: finding them is part of the item. Q16's angles are 54°, 84° and 42° as drawn.

Partial-credit rules

- **Questions 1-10 (2 points).** Correct letter: 2. Otherwise 0. No partial credit.
- **Questions 11-20 (5 points).** Correct letter: 5. On items with a method box, when the letter is wrong or blank: 3 points for a correct representation with a single arithmetic slip; 2 points for a correct representation that stops or goes wrong later; 0 otherwise. Items without a method box: 5 or 0.
- **Questions 21-25 (6 points).** Correct answer: 6. Otherwise: 2 points for a correct representation, 4 points for a correct representation plus the key intermediate result below; 0 otherwise.
- Score the persistence rubric on its own sheet. Never add or subtract persistence points here.

Q	Correct representation (2)	Adds to 3 (5-pt) / 4 (6-pt)
11	A correct two-step chain (sale price 60, then tax on 60) or $80 \times 0.75 \times 1.13$.	Chain correct but one multiplication slipped.
12	A bar model or ratio table with 8 equal parts totalling 48.	Part size 6 found but final difference slipped.
14	A table of squares vs toothpicks showing +3, or the rule $3n + 1$.	Rule correct, final evaluation slipped.
15	The figure split into two rectangles, or the 10×8 rectangle with the missing 4×3 corner marked.	Correct decomposition, one product or sum slipped.
18	An equation such as $5 \div \text{total} = 1/3$, or $\text{total} = 15$ stated.	Total 15 found, subtraction slipped.
19	A Venn diagram or the three counts 33, 20 and 6.	Counts correct, combination slipped.
21	A bar/ratio model with parts $3k$ and $4k$, or $3k + 6 = 4k$.	$k = 6$ (or 18 boys and 24 girls before).
22	Numbers written as $n - 4$, $n - 2$, n , $n + 2$, $n + 4$ (or n , $n + 2$, ...).	The middle number is 34.
23	A sketch or layer view marking corner, edge and face cubes.	2 two-face cubes on each edge.
24	A 6×6 outcome grid or organised list of ordered pairs.	5 favourable outcomes identified.
25	An organised list or table by hundreds digit.	At least three cases counted correctly (e.g. 1, 2, 3).

Per-item teacher notes

Q	Answer	Concept tested	Likely error	Faster method
1	(D) 27	Fraction of a quantity	(A) 9 — found $1/4$ only	$36 \div 4 = 9$, then $\times 3$.
2	(B) 12	Percent of a quantity	(E) 65 — subtracted 15 from 80	10% is 8, 5% is 4; $8 + 4$.
3	(C) 9	Solving a one-step-then-two-step equation	(D) 12.5 — added 7 instead of subtracting: $50 \div 4$	Undo: $43 - 7 = 36$, $36 \div 4 = 9$.
4	(D) 68	Arithmetic sequence: nth term	(E) 75 — used $5 + 10 \times 7$ (one step too many)	10th term = first + 9 steps = $5 + 63$.
5	(C) 38 cm	Perimeter of a rectangle	(D) 84 cm — found the area	$2 \times (12 + 7)$.
6	(C) 65°	Angle sum of a triangle	(D) 115° — added the two angles and stopped	$180 - 48 - 67 = 180 - 115$.
7	(D) 14	Mean of a data set	(B) 12 — gave the median	Sum 70, $\div 5$.

Q	Answer	Concept tested	Likely error	Faster method
8	(A) 59	Primes; divisibility by 3 and 7	(E) 91 — looks prime, but $91 = 7 \times 13$	Digit sums 12, 6, 12 show 39, 51, 57 are multiples of 3; $91 = 7 \times 13$.
9	(E) 24	Lowest common multiple	(A) 2 — gave the greatest common factor	List multiples of 8 until one is a multiple of 6: 8, 16, 24.
10	(A) Ben	Logical deduction from constraints	(E) — did not combine the two clues	Cat is Cy's; Ben is not dog, so Ben has the fish.
11	(B) \$67.80	Successive percent change	(C) \$70.40 — added -25% and $+13\%$ to get -12% of \$80	$80 \times 0.75 = 60$; $60 \times 1.13 = 67.80$.
12	(B) 12	Ratio with a fixed total (parts of a whole)	(A) 6 — found one part ($48 \div 8$) and stopped	8 parts = 48, so a part is 6; blue – red = 2 parts = 12.
13	(C) $\frac{1}{2}$	Fraction of a remainder ('of what')	(B) $\frac{5}{12}$ — took $\frac{1}{4}$ of the whole, not of the rest	Left after Mia: $\frac{2}{3}$; Leo leaves $\frac{3}{4}$ of that: $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$.
14	(B) 61	Linear pattern from a table	(D) 80 — used 4 per square, ignoring shared sides	First square 4, each new square adds 3: $4 + 19 \times 3$.
15	(C) 68	Composite area; finding a missing side	(E) 80 — used the full 10×8 rectangle	$10 \times 8 - (10 - 6) \times (8 - 5) = 80 - 12$.
16	(B) 42	Angles on a straight line; one-variable equation	(C) 63 — solved $2x + 54 = 180$	$3x = 180 - 54 = 126$.
17	(C) 5 cm	Volume of a prism (reverse)	(E) 45 cm — divided by 8 only	Base area 72; $360 \div 72$.
18	(D) 6	Probability as a fraction of the total; reverse	(E) 15 — gave the total, not the green count	Blue is $\frac{1}{3}$ of the total, so total = 15; green = $15 - 9$.
19	(C) 47	Inclusion-exclusion counting	(D) 53 — counted multiples of 15 twice	$33 + 20 - 6$.
20	(D) Dee	Ordering from constraints	(C) Cal — placed Cal before Dee	Chain: Ari, Bo, Dee, then Cal right after Dee.
21	48	Ratio change when one part changes	42 — the total before the 6 boys joined	The gap of 1 part is 6, so 4 parts = 24 girls; now $24 + 24$.
22	38	Consecutive numbers: sum = count $\times$ middle	34 — gave the middle number	Middle = $170 \div 5 = 34$; largest = $34 + 4$.
23	24	Spatial counting on a cube's edges	8 — counted the corner cubes (three faces)	Each of the 12 edges has $4 - 2 = 2$ middle cubes: 12×2 .
24	$\frac{5}{36}$	Outcome space for two dice	$\frac{5}{11}$ — counted sums as equally likely	Pairs (2,6), (3,5), (4,4), (5,3), (6,2): 5 of 36.
25	15	Organised casework; completeness	10 or 21 — missed or double-counted cases (e.g. allowed a leading 0)	Hundreds digit 5, 4, 3, 2, 1 gives $1 + 2 + 3 + 4 + 5$ cases.